SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

***CO1:** Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Total Marks:25**

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage (Marks)	Level 4 – Exemplary (Full Marks)	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	35 Marks	Demonstrates clear, complete, and in-depth understanding of OS concepts (types, structure, system calls)	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answers	25 Marks	All answers are correct, precise, and relevant	Mostly correct with minor errors	Partially correct with noticeable errors	Mostly incorrect or irrelevant
Application of Knowledge	15 Marks	Effectively applies concepts to scenarios/questions	Applies with minor mistakes	Limited or partial application	No meaningful application
Completeness of Response	15 Marks	All parts of each question are fully answered	Most parts covered with small omissions	Some parts missing	Incomplete answers
Clarity & Presentation	10 Marks	Clear, well-structured, and neatly presented answers	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

CSA04 - Operating Systems

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ASSESSMENT TOOL - I

SHORT ANSWER TEST.

1) a.) Types of Operating System:

① Multi-user operating System:

* Allow multiple users to access the System

Ex: Linux, UNIX.

② Multitasking operating System:

* Executes multiple processes at the same time by CPU time.

* Improves system efficiency.

③ Multiprocessing operating System:

* Uses two or more CPUs/Cores to execute processes.

* Increase performance.

④ Time-sharing Operating System:

* Allocates CPU time equally among users and processes.

* Ensures quick response for all users.

b.) Structure of an Operating System.

Users.

↓

Application Program

↓

System Calls.

↓

operating system.

- → Process Management
- → memory Management
- → File Management
- → I/O Device Management
- → CPU Schedular.

↓

Hardware (CPU, memory, Disk, I/O)

c.) sequence of System Calls to Create and Execute a New Process.

start

↓

fork() process

↓

exec()
a new program
↓
wait()
child to Complete
↓
exit()
terminates
↓
End.

2)
a.) Categories of System Calls.
File and Device
File Management System Calls.
* open()
* read()
* write()
* close()
* Create()
* delete()

Device Management System Calls:
* read()
* write()
* ioctl()
* Open()
* close()

b) Layered Structure of File operations

User

↓

Application Program

↓

System Call Interface

↓

File System

↓

Device Drivers

↓

Storage Hardware

Explakation:

* User requests a file operation.
* Application invokes a system Call.
* File System processes the request.
* Device driver communicates with the storage device.
* Hardware performs the operation.

c.) Pseudo Code / C- based System Call Sequence.

Pseudo Code :

START

Open file

If file opened successfully.

Read data from file

Display data

close file

Else

Display "File Cannot be opened".

STOP.

C Program

```
# include < stdio.h >
# include < fcntl.h >
# include <unistd.h >
int main () {
    char buffer [100];
    int fd = open ("Sample.txt", O_RDONLY);
      if (fd != -1) {
          read(fd, buffer, Size of (buffer));
              printf("%s", buffer);
              close (fd);
          } else {
              printf("Unable to open file.");
          }
          return 0;
    }
```